El Niño–Southern Oscillation (ENSO)

Coupled dynamics, monitoring, and global impacts

NCSU | MEA 458

Learning goals

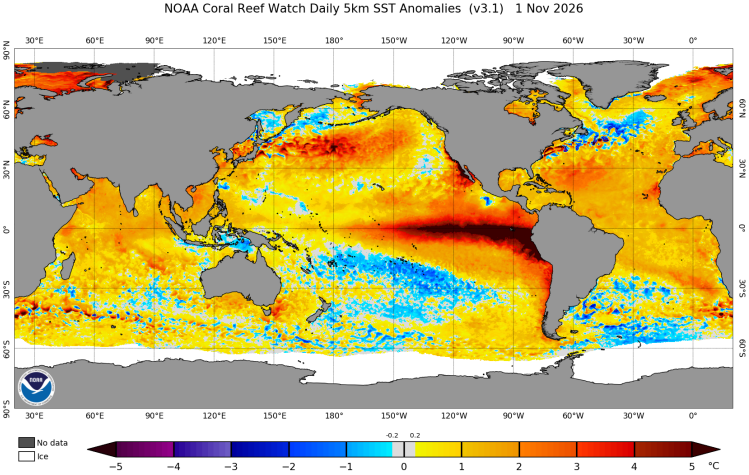
1. Distinguish the three ENSO phases and describe the mean tropical Pacific.
2. Explain Bjerknes feedback and why ENSO reverses phase.
3. Interpret SST, pressure, and subsurface ENSO indices.
4. Trace tropical heating anomalies to global climate impacts.

What is ENSO?

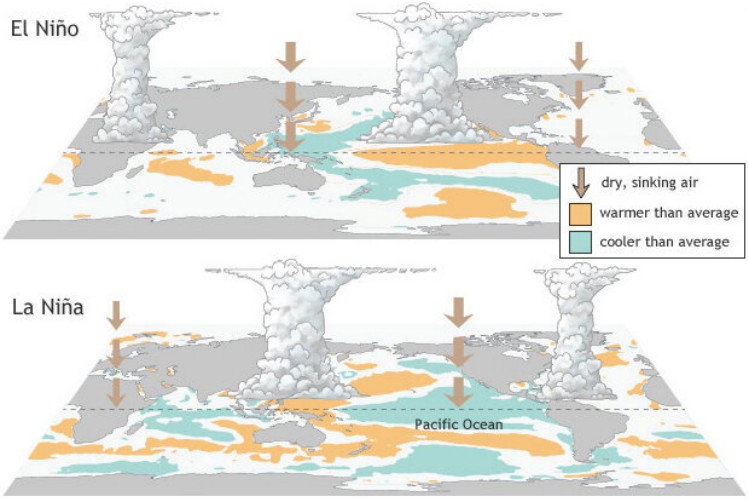
- **El Niño**: sustained anomalous warming of the central/eastern equatorial Pacific, coupled to atmospheric changes.
- **La Niña**: sustained anomalous cooling and generally strengthened trade winds.
- **Southern Oscillation**: large-scale seesaw in tropical Pacific–Indian Ocean sea-level pressure.
- ENSO is irregular (roughly 2–7 years), not a perfectly periodic oscillation.

Source: McPhaden et al. (2006); Timmermann et al. (2018)

Current State



EL Nino Vs La Nina



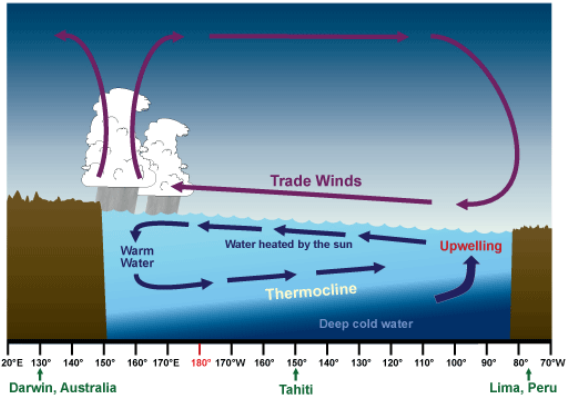
Source https://www.climate.gov/news-features/blogs/enso/what-el-nino-southern-oscillation-enso-nutshell

Thermocline

A thermocline is the transition layer between warmer mixed water at the ocean's surface and cooler deep water below (NOAA)

Normal tropical Pacific (ENSO Neutral)

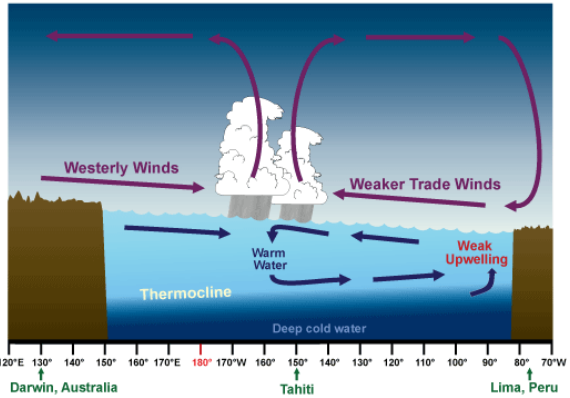
Source: https://www.noaa.gov/jetstream/tropical/enso/enso-patterns



1. Normally, sea surface temperature is about 14°F (8°C) higher in the Western Pacific than the waters off South America.
2. This is due to the trade winds blowing from east to west along the equator, allowing the upwelling of cold, nutrient rich water from deeper levels off the northwest coast of South America.
3. These same trade winds push water west, which piles higher in the Western Pacific. The average sea level height is about 1½ feet (46 cm) higher near Indonesia than Peru.
4. The trade winds, in piling up water in the Western Pacific, make a deep 450-foot (150 meter) warm layer in the west that pushes the thermocline down while it rises in the east.
5. The shallow 90-foot (30 meter) eastern thermocline allows the winds to pull up water from below, water that is generally much richer in nutrients than the surface layer.

El Nino Conditions

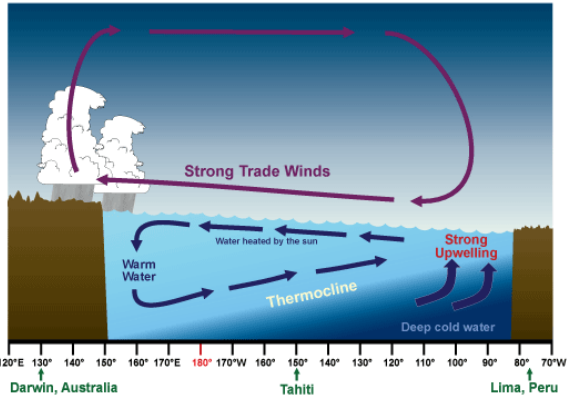
Source: https://www.noaa.gov/jetstream/tropical/enso/enso-patterns



1. When the air pressure patterns in the South Pacific reverse direction (the air pressure at Darwin, Australia is higher than at Tahiti), the trade winds decrease in strength (and can also reverse direction).
2. As a result, the normal flow of water away from South America decreases, and ocean water piles up off South America. This pushes the thermocline deeper and decreases upwelling.
3. With a deeper thermocline and decreased westward transport of water, the sea surface temperature increases to greater than normal in the Eastern Pacific. This is the warm phase of ENSO, called El Niño.
4. The net result is a shift in the prevailing rain pattern from the normal, with the Western Pacific now becoming relatively dry and rainfall increasing in the Central Pacific.

La Nina Conditions

Source: https://www.noaa.gov/jetstream/tropical/enso/enso-patterns



1. There are also occasions when the trade winds that blow west across the tropical Pacific are stronger than normal. These winds pile up warm surface water in the West Pacific. This leads to increased upwelling off South America and leading to lower than normal sea surface temperatures.
2. The prevailing rain pattern also shifts farther west than normal.
3. This is the cool phase of ENSO called La Niña.
4. What some may find surprising is that these changes in sea surface temperatures are not large, plus or minus 6°F (3°C) and generally much less, yet they have a large impact on the weather.

Activity 1: predict the response

Think–pair–share. Suppose the equatorial trade winds weaken for several weeks. Predict the sign of change in (a) eastern Pacific thermocline depth, (b) eastern SST, (c) central Pacific convection. Explain the links rather than listing answers.

El Niño versus La Niña

Feature	El Niño	La Niña
Trade winds	Usually weaker	Usually stronger
East Pacific SST	Warmer	Cooler
East thermocline	Deeper	Shallower
Central/east convection	Enhanced/shifted east	Often suppressed
Walker circulation	Weakened/reorganized	Strengthened/reorganized

Source: McPhaden et al. (2006); Timmermann et al. (2018)

Bjerknes Feedback: A Positive Ocean–Atmosphere Feedback

The **Bjerknes feedback** is a positive feedback mechanism that amplifies SST anomalies in the equatorial Pacific.

During El Niño:

1. **Initial warming:** SSTs increase in the central and eastern equatorial Pacific.
2. **Atmospheric response:** The east–west SST gradient weakens, reducing the Walker circulation and easterly trade winds.
3. **Oceanic response:** Weaker trades deepen the eastern Pacific thermocline and reduce the supply of cold subsurface water to the surface.
4. **Amplification:** Reduced cooling leads to further SST warming, reinforcing the initial anomaly.

Positive Feedback Loop

Warmer eastern Pacific $\rightarrow$ Weaker trades $\rightarrow$ Deeper eastern thermocline
$\rightarrow$ Reduced upwelling cooling $\rightarrow$ Further warming

Bjerknes positive feedback

Key point: Bjerknes feedback explains the *growth* of ENSO anomalies, but not their eventual termination or reversal.

Source: Bjerknes feedback summarized in Timmermann et al. (2018); original schematic

Why positive feedback alone is insufficient

- Positive feedback explains *growth*, not termination or phase reversal.
- Ocean dynamics redistribute warm water volume across the equatorial Pacific.
- Wind-driven Kelvin and Rossby waves adjust thermocline depth with delays.
- Seasonal modulation and episodic westerly wind bursts complicate event evolution.

Source: Timmermann et al. (2018); Vialard (2025)

ENSO Theory: The Delayed Oscillator

Central idea: Bjerknes feedback amplifies ENSO anomalies, while oceanic waves provide a **delayed negative feedback** that eventually reverses them.

Evolution during El Niño:

1. **Initial warming:** Eastern Pacific SST warming weakens the easterly trade winds.
2. **Wave generation:** Anomalous westerly winds generate:
   - Eastward-propagating downwelling Kelvin waves.
   - Westward-propagating upwelling Rossby waves.
3. **Western boundary reflection:** Rossby waves reach the western Pacific boundary and partially reflect as eastward-propagating upwelling Kelvin waves.
4. **Delayed cooling:** Reflected Kelvin waves shoal the eastern Pacific thermocline, increasing cold-water entrainment and weakening El Niño.

Key Concept

The negative feedback is delayed by the time required for oceanic waves to propagate across the Pacific.

Delayed Oscillator: The core idea

In one sentence

The same wind change that strengthens an El Niño also sends out a slow ocean signal that comes back months later and reverses it.

- Growth: fast, local, positive feedback
- Termination: slow, remote, negative feedback that arrives *late*
- A push forward plus a delayed push backward $\Rightarrow$ oscillation

Delayed Oscillator: Setting the scene: the thermocline

- The tropical Pacific has a warm surface layer over cold deep water. The boundary is the **thermocline**.
- Trade winds pile warm water up in the west:
  - **West** (Indonesia): thermocline deep ($\sim$150 m), surface warm
  - **East** (Peru): thermocline shallow ($\sim$50 m), cold water upwelled easily
- Moving the eastern thermocline up or down changes eastern sea surface temperature (SST).
- Ocean waves are what move it.

Two effects, two wave types

What the wave does

- **Downwelling**: pushes the thermocline down $\Rightarrow$ warming
- **Upwelling**: lifts the thermocline up $\Rightarrow$ cooling

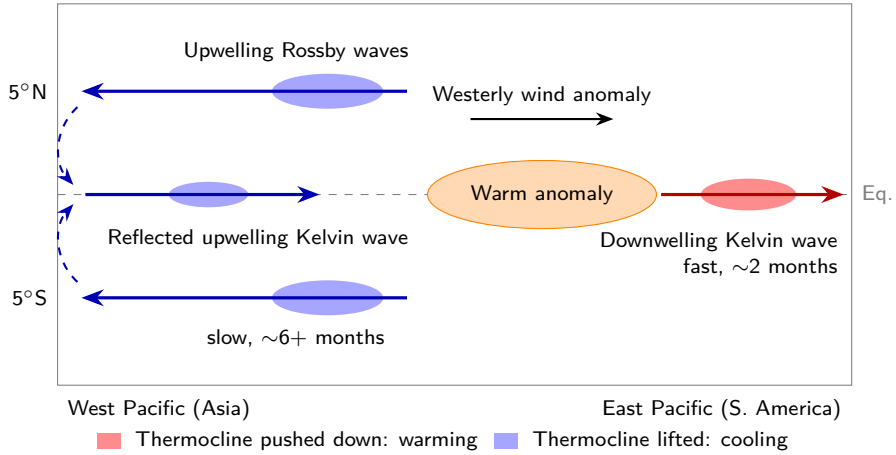
How it travels

	Kelvin	Rossby
Location	On the equator	$\sim 5^\circ$N and S
Direction	East only	West only
Speed	$\sim$2.5–3 m/s	$\sim$1/3 of Kelvin
Crossing	$\sim$2 months	$\sim$6+ months

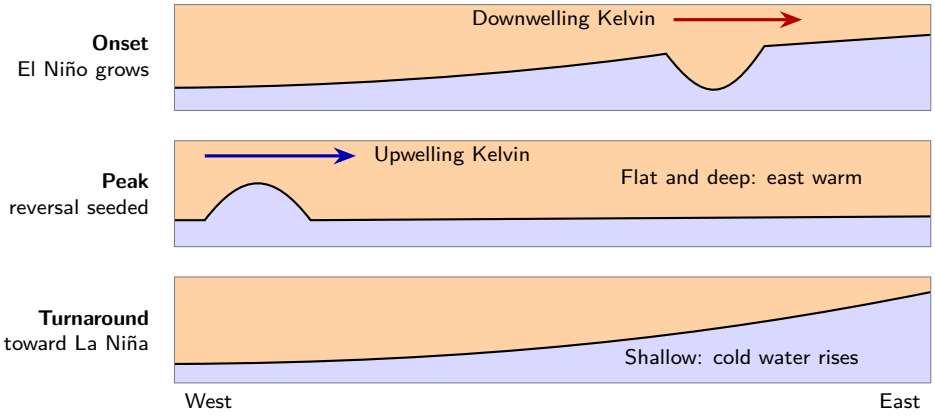
The sequence, step by step

1. A **warm anomaly** appears in the central or eastern Pacific.
2. Trade winds weaken: a **westerly wind anomaly** over the warm patch.
3. The wind launches two signals at once:
   - East: a **downwelling Kelvin wave** deepens the thermocline $\Rightarrow$ El Niño grows
   - West: **upwelling Rossby waves** head slowly toward Asia
4. **Reflection**: the Rossby waves bounce off the western boundary as an **upwelling Kelvin wave**.
5. **Reversal**: it reaches the east, lifts the thermocline, cold water returns $\Rightarrow$ El Niño ends, often La Niña begins.

Schematic 1: map view of the waves



Schematic 2: the thermocline along the equator



Orange: warm surface layer Blue: cold deep water Black line: thermocline

Why “delayed”?

- The terminating signal is born *at the same time* as the growth, but takes a long detour: west slowly (Rossby), then back east (Kelvin).
- That travel time is the delay δ.

Suarez & Schopf (1988); Battisti & Hirst (1989)

$$\frac{dT}{dt} = \underbrace{aT}_{\text{local growth}} - \underbrace{b\,T(t-\delta)}_{\text{delayed negative feedback}} - \underbrace{c\,T^3}_{\text{damping}}$$

- T: eastern Pacific SST anomaly
- With realistic wave speeds, the period comes out at about 3–5 years, close to observed ENSO.

Intuition and limitations

Analogy

Steering a boat when the feedback from the rudder arrives months late: you keep oversteering, and the boat swings back and forth.

Limitations

- Real ENSO is irregular. Westerly wind bursts, seasonality and noise matter, and the simple delayed oscillator is too regular.
- Eastern-boundary reflection and the role of the total warm-water volume are better captured by the **recharge oscillator** (Jin 1997).
- The two are best seen as complementary views of the same physics.

ENSO Theory: Recharge–Discharge Oscillator

Central idea: ENSO involves the accumulation and depletion of upper-ocean heat content in the equatorial Pacific.

The ENSO cycle:

1. **Recharge phase:** Anomalous ocean circulation transports warm upper-ocean water toward the equatorial region.
   - Equatorial heat content increases.
   - Basin-mean equatorial thermocline deepens.
2. **El Niño development:** Deepened thermocline and weakened trade winds favor eastern Pacific warming through Bjerknes feedback.
3. **Discharge phase:** Wind-driven ocean circulation redistributes warm upper-ocean water away from the equator.
   - Equatorial heat content decreases.
   - Basin-mean equatorial thermocline shoals.
4. **Transition to La Niña:** Reduced equatorial heat content favors eastern Pacific cooling and strengthened trade winds.

Activity 2: why does El Niño end?

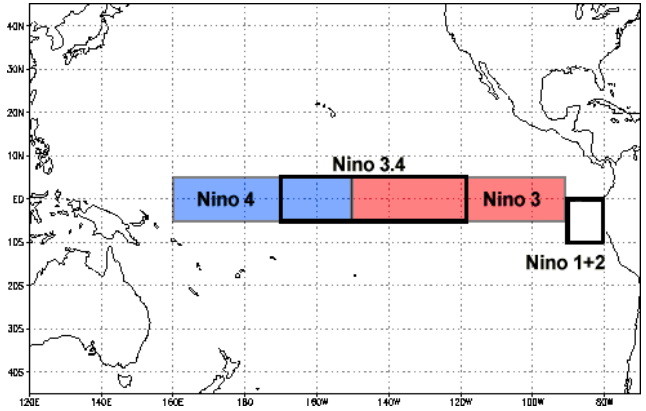
In pairs: Bjerknes feedback amplifies El Niño. Identify one physical process that could eventually reverse the SST anomaly. Distinguish *amplification* from *delayed negative feedback*.

ENSO diversity and predictability (2 min)

- Events differ in amplitude, longitudinal SST structure, and onset.
- Eastern-Pacific and central-Pacific events need not produce identical teleconnections.
- Seasonal phase locking: many events peak during boreal winter.
- Spring predictability barrier: forecast skill often drops across boreal spring.

Source: Timmermann et al. (2018)

Where do we measure ENSO? (2 min)



Niño 3.4: 5°S–5°N, 170°W–120°W. The rectangles are schematic coordinate regions, not a coastline map.

Source: NOAA CPC / PSL; original schematic

ENSO monitoring: not SST alone

- **Niño 3.4 / ONI**: equatorial SST anomaly; ONI is a 3-month running mean.
- **RONI**: relative Niño-3.4 SST index adjusted for tropical-mean SST anomaly and rescaled; operational at NOAA since February 2026.
- **SOI**: standardized Tahiti–Darwin pressure difference; generally negative in El Niño.
- **MEI**: combines oceanic and atmospheric indicators.
- Subsurface heat content, thermocline depth, trade winds, OLR, and TAO/Argo observations diagnose evolution.

Source: NOAA (2026); NOAA PSL ENSO dashboard

Activity 3: diagnose an event (3 min)

Hypothetical observations: Niño-3.4 SST $+1.2^{\circ}$C; SOI negative; easterly trades weakened; eastern thermocline deeper than normal.

1. Which ENSO phase is consistent with these observations?
2. Why does agreement across atmosphere and ocean increase confidence?
3. Can a single monthly SST anomaly establish a sustained ENSO event?

Global impacts: the physical pathway

Anomalous tropical convective heating excites large-scale circulation changes and Rossby-wave trains. Teleconnections alter storm tracks and regional hydroclimate; they are probabilistic, not deterministic.

Source: McPhaden et al. (2006); Timmermann et al. (2018); original schematic

Global impacts: regional examples

- **Maritime Continent / Australia**: increased drought risk in many El Niño events.
- **Peru / Ecuador**: some eastern-Pacific El Niño events bring extreme coastal rainfall.
- **North America**: winter jet/storm-track shifts often favor wetter southern US conditions during El Niño.
- **South Asia**: ENSO affects monsoon probabilities, but the relationship is nonstationary and far from one-to-one.

Impacts. See https://www.noaa.gov/jetstream/enso¡*mpacts*

ENSO and tropical cyclones

- El Niño typically increases upper-tropospheric westerlies and vertical wind shear over the tropical Atlantic, suppressing Atlantic TC activity on average.
- La Niña generally favors reduced Atlantic shear and more Atlantic TC activity.
- In the western North Pacific, ENSO affects genesis longitude, track distribution, and lifetime.
- ENSO shifts *probabilities*; it does not determine any individual storm.

Source: Camargo et al. (2007); Timmermann et al. (2018)

Synthesis discussion

A sustained central-Pacific warm anomaly develops.

1. Sketch the likely wind, thermocline, and convection response.
2. Explain why positive feedback does not imply perpetual warming.
3. Select two monitoring variables beyond SST.
4. Explain one plausible pathway from tropical heating to a midlatitude climate anomaly.

Selected published references

- McPhaden, M. J., S. E. Zebiak, and M. H. Glantz (2006): ENSO as an integrating concept in Earth science. *Science*, 314, 1740–1745. doi:10.1126/science.1132588.
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